University of Florida, Department of Mathematics

Analysis PhD Exam, August 2025

Write solutions in a neat and logical fashion, giving complete reasons for all steps. Do six of the eight problems.

1. Do two. Short answer, you do not need to give a detailed proof, just a brief explanation.

(a) If $a_1, \dots, a_n$ are non-negative numbers, must it be the case that

$$\left(\sum_{j=1}^{n} a_j\right)^5 \leq n^4 \sum_{j=1}^{n} a_j^5?$$

(b) Define $f \in L^1(\mathbb{R})$ by $f(x) = e^{-x^4}\chi_{[1,\infty)}(x)$, where χ denotes indicator function. Is the Fourier transform of f in $L^1(\mathbb{R})$?

(c) If $g, h \in L^1(\mu)$, when are $g\,d\mu$ and $h\,d\mu$ mutually singular?

2. Suppose $(X, \mathcal{M})$ is a measurable space. Show, if $E \in \mathcal{M} \otimes \mathcal{M}$, then

$$\{x \in X : (x, x) \in E\} \in \mathcal{M}.$$

3. This problem has two parts.

(a) Let $f_n(x) = x^{-n}$ for $n \geq 2$. Does the numerical sequence

$$c_n = \int_1^\infty e^{inx} f_n(x)\, dx$$

converge?

(b) Give an example, if possible, where strict inequality holds in Fatou's Lemma.

4. Suppose X is a Banach space, $Y, Z \subseteq X$ are (closed) subspaces and $Y \cap Z = \{0\}$. Prove or disprove: there is a $C > 0$ such that

$$C(\|y\| + \|z\|) \leq \|y + z\|$$

for all $y \in Y$ and $z \in Z$.

5. Highlighting the key points, outline a proof of the following statement. If a sequence of functions converges in L^1, then it has a subsequence that converges pointwise almost everywhere.

6. Suppose $(X, \mathcal{M}, \mu)$ is a measure space, $1 \leq p, q \leq \infty$ and $h : X \to \mathbb{C}$ is measurable. Show, if $hf \in L^q(\mu)$ for each $f \in L^p(\mu)$, then the map

$$L^p(\mu) \ni f \mapsto hf \in L^q(\mu)$$

is continuous.

7. Define the Hardy–Littlewood maximal function of a Lebesgue measurable function $f : \mathbb{R} \to \mathbb{R}$. State the Hardy–Littlewood maximal inequality. Show, if $f \in L^1(\mathbb{R})$, then f^* (the Hardy–Littlewood maximal function of f) is finite almost everywhere.

8. Do two. Short answer, you do not need to give a detailed proof, just a brief explanation.

(a) Does there exist a bounded linear functional $\lambda : L^\infty(\mathbb{R}) \to \mathbb{C}$ for which there does not exist a $g \in L^1(\mathbb{R})$ such that $\lambda(f) = \int fg\, dx$?

(b) Does the norm in $\ell^1(\mathbb{N})$ arise from an inner product?

(c) Can a sigma algebra be bijective to $\mathbb{N}$?